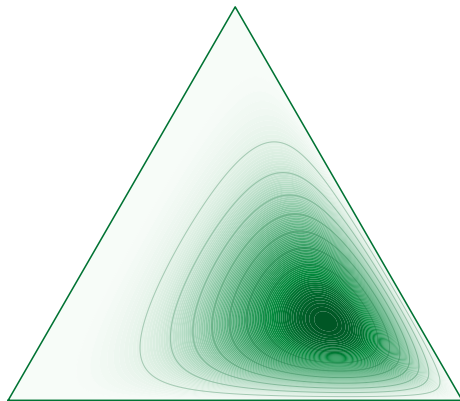


Evidential Bayesian Neural Networks for Uncertainty-Aware Classification

John-Pierre Weideman

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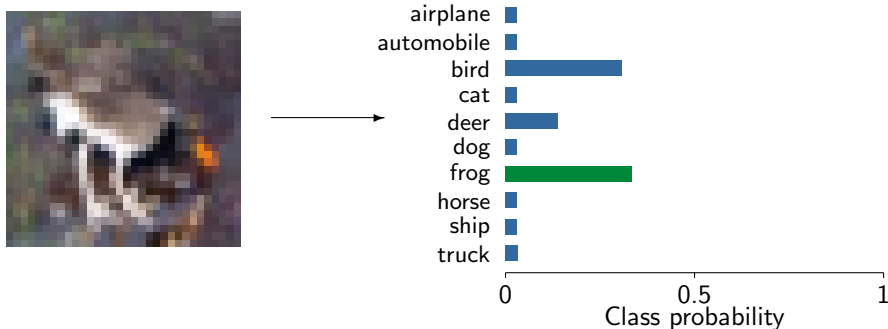
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Why Uncertainty-Aware Modelling Matters

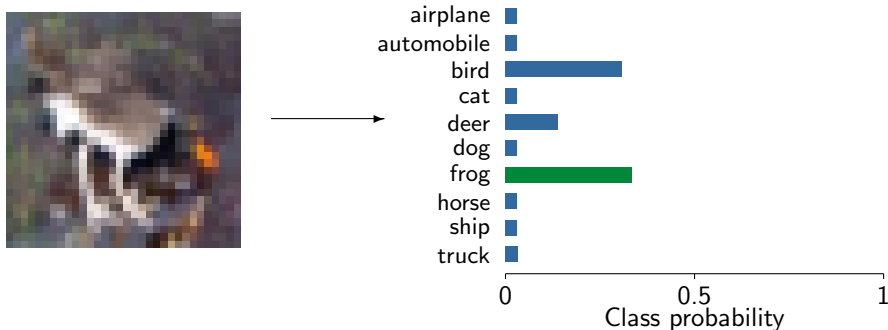


- Estimating uncertainty can support safer decisions.
- Especially important when errors have serious consequences.
 - ▶ Autonomous driving, medical, financial, legal, etc.

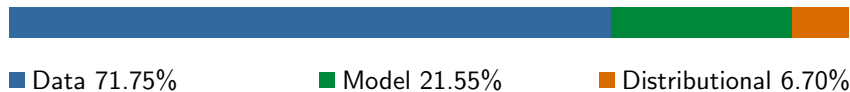
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- Predictive uncertainty* is uncertainty about predicted class label of an input.
 - Does not by itself guide decisions about how to act.
 - Where does the predictive uncertainty come from?



- Different uncertainty sources require different actions.
- **Aleatoric (data) uncertainty** irreducible uncertainty from complexity in the data.
 - ▶ Defer the decision if high risk.
- **Epistemic (model) uncertainty** reducible uncertainty in the estimated network weights.
 - ▶ Collect more relevant training examples.
- **Distributional uncertainty** is uncertainty about the class probabilities.
 - ▶ If inputs differ from training data, train on more representative examples.

- **Bayesian neural networks**

- ▶ Model uncertainty over network weights.
- ▶ Aleatoric and epistemic uncertainty.

- **Deterministic Dirichlet networks**

- ▶ Model a distribution over class probabilities
- ▶ Aleatoric and distributional uncertainty.

- Neither decomposes into all three uncertainty terms.

- Model that represents aleatoric, distributional, and epistemic uncertainty.
 - ▶ Combine **Bayesian** and **Dirichlet neural network**.

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- C classes, input x , label $y \in \{1, \dots, C\}$, training data $D = \{(x_i, y_i)\}_{i=1}^N$.
- The network with weights w outputs class-probability vector,

$$f(x; w) = (f_1(x; w), \dots, f_C(x; w)), \quad f_c(x; w) \geq 0, \quad \sum_{c=1}^C f_c(x; w) = 1.$$

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- Categorical likelihood and weight prior define the posterior over weights.

$$p(D | w) = \prod_{i=1}^N \text{Cat}(y_i | f(x_i; w)) = \prod_{i=1}^N f_{y_i}(x_i; w),$$

$$\underbrace{p(w | D)}_{\text{posterior}} \propto \underbrace{p(D | w)}_{\text{likelihood}} \underbrace{p(w)}_{\text{prior}}.$$

- Random class-probability vector $\pi = (\pi_1, \dots, \pi_C)$, $\pi_c \geq 0 \ \forall c$, $\sum_{c=1}^C \pi_c = 1$.
- π is Dirichlet distributed with concentration parameters $\alpha = (\alpha_1, \dots, \alpha_C)$, $\alpha_c > 0 \ \forall c$,
$$p(\pi \mid \alpha) = \text{Dir}(\pi \mid \alpha).$$

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$$p(\pi \mid \alpha) = \text{Dir}(\pi \mid \alpha).$$
- Mean vector and total concentration parameter,

$$m = \mathbb{E}[\pi] = \frac{\alpha}{\alpha_0}, \quad \alpha_0 = \sum_{c=1}^C \alpha_c.$$

- Dirichlet-categorical network outputs $\alpha(x; w)$.

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$

$$p(y \mid \pi) = \text{Cat}(y \mid \pi) = \pi_y.$$

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- Marginalise π to obtain label probabilities for input x .

$$p(y \mid x, w) = \int p(y \mid \pi) p(\pi \mid x, w) d\pi$$
$$= \mathbb{E}_{\text{Dir}(\pi \mid \alpha(x; w))}[\pi_y] = \frac{\alpha_y(x; w)}{\alpha_0(x; w)} = m_y(x; w).$$

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- **Evidential deep learning (EDL)** parameterises the concentrations through evidence $e_c(x; w) > 0$, with $\alpha_c(x; w) = e_c(x; w) + 1$.
 - ▶ Use subjective logic to interpret evidence as support for classes.
 - ▶ Zero evidence corresponds to uniform Dirichlet.

- *Categorical BNN*: Point mass $P(\pi = f(x; w) | x, w) = 1$.
 - ▶ Zero distributional uncertainty.
- *Deterministic Dirichlet network*: Point mass $P(w = \hat{w}) = 1$.
 - ▶ Zero epistemic uncertainty.
- Combining **weight posterior** with a **Dirichlet distribution** over class probabilities represents all three uncertainty terms.

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- **One observed label** per input.

- **Network outputs** $\alpha(x; w) = e(x; w) + 1$,

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$

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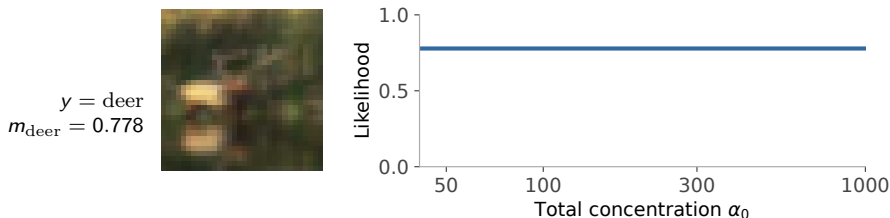
- Assuming conditionally independent labels, **likelihood** is

$$p(D \mid w) = \prod_{i=1}^N p(y_i \mid x_i, w) = \prod_{i=1}^N m_{y_i}(x_i; w).$$

$$p(D | w) = \prod_{i=1}^N m_{y_i}(x_i; w) = \prod_{i=1}^N \frac{\alpha_{y_i}(x_i; w)}{\alpha_0(x_i; w)}$$

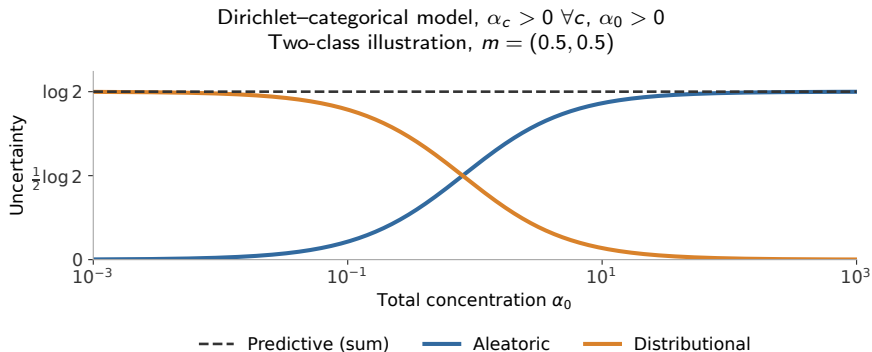
- Likelihood **does not inform total concentration**

- ▶ Does not favour one total concentration over another, at fixed mean vectors.



Total Concentration Controls the Uncertainty Allocation

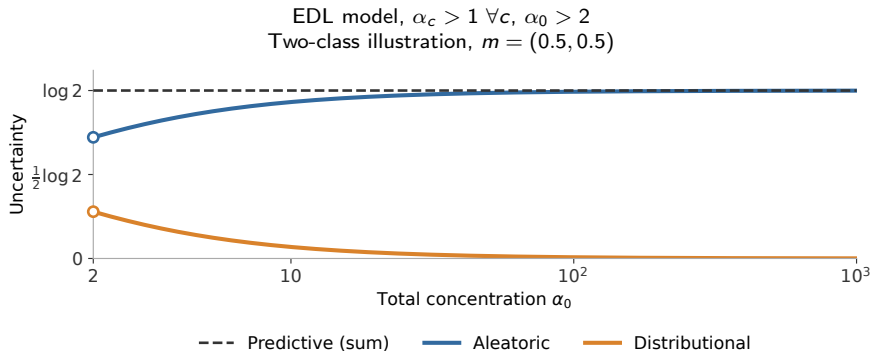
- At fixed mean vector, **total concentration** controls **aleatoric–distributional** allocation.



- Single labels **cannot inform** this uncertainty allocation.

A Limitation of the Evidence Parameterisation

- EDL **restricts** the aleatoric–distributional split.
 - ▶ Fully distributional allocation cannot be approached.



- Would **less restrictive** parameterisation $\alpha_c > 0 \forall c$ be preferable?

Prior on Total Concentration

- At fixed mean vectors, single labels **do not inform total concentration**.
 - ▶ And therefore not the uncertainty allocation.

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- State the **total concentration assumption** through prior,

$$\tilde{p}(w) \propto p(w) \prod_{i=1}^N \text{Gamma}(\alpha_0(x_i; w) \mid \kappa, \beta).$$

- ▶ Shape $\kappa > 1$, rate $\beta > 0$.
- ▶ Prior mode $(\kappa - 1)/\beta > C$ and standard deviation $\sqrt{\kappa}/\beta$.

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 - ▶ Prior mode $(\kappa - 1)/\beta > C$ and standard deviation $\sqrt{\kappa}/\beta$.
- Prior $\tilde{p}(w)$ and categorical likelihood give weight posterior,

$$p(w \mid D) \propto \tilde{p}(w) \prod_{i=1}^N m_{y_i}(x_i; w).$$

Uncertainty Allocation Depends on Prior Assumption

- Vary Gamma **prior mode** (level of total concentration).

Prior mode	Predictive entropy	Aleatoric %	Distributional %	Epistemic %
Fashion-MNIST				
30	1.336 (1.333–1.341)	90.36	9.39	0.25
58	0.934 (0.920–0.958)	92.51	7.07	0.42
100	0.974 (0.934–1.008)	95.42	4.13	0.45
300	1.947 (1.922–1.959)	99.12	0.77	0.11
CIFAR-10				
30	1.414 (1.399–1.436)	89.24	8.98	1.78
100	0.818 (0.813–0.822)	91.11	4.79	4.10
300	1.733 (1.728–1.739)	98.07	0.87	1.06

- Higher prior mode $\Rightarrow$ **larger aleatoric** and **smaller distributional** shares.
 - ▶ Uncertainty allocation depends on the **prior assumption**.
 - ▶ Prior assumption moves allocation as theory predicts.

- Compare predictive performance of Cat-eBNN against Cat-BNN.

model	accuracy	NLL
Fashion-MNIST		
Cat-BNN	0.921 (0.917–0.924)	0.221 (0.211–0.230)
Cat-eBNN	0.913 (0.908–0.916)	0.431 (0.410–0.458)
CIFAR-10		
Cat-BNN	0.856 (0.850–0.859)	0.426 (0.418–0.441)
Cat-eBNN	0.867 (0.866–0.868)	0.470 (0.461–0.476)

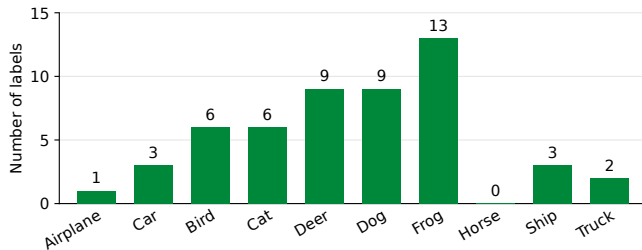
- Competitive, but **no consistent predictive improvement**.

- Cat-eBNN allow decomposition into all three uncertainty terms.
- But allocation reflects prior assumption, not something model learns from the data.
- Not clear practical value.
- **Can data inform uncertainty allocation?**

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- **Repeated labels** are multiple observed labels for same input.



- CIFAR-10H example with 52 human labels.
- A single selected label **hides observed disagreement**.

- **Repeated labels** per input give counts $n = (n_1, \dots, n_C)$, with $R = \sum_c n_c$.

- **Network outputs** $\alpha(x; w) = e(x; w) + 1$.

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$

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- For count data $D_{\text{count}} = \{(x_i, n_i)\}_{i=1}^N$, the **likelihood** is

$$p(D_{\text{count}} \mid w) = \prod_{i=1}^N p(n_i \mid x_i, w) = \prod_{i=1}^N \text{DM}(n_i \mid R_i, \alpha(x_i; w)).$$

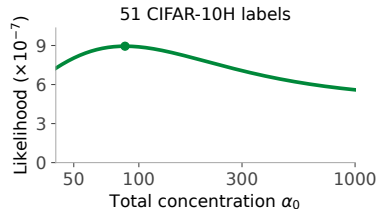
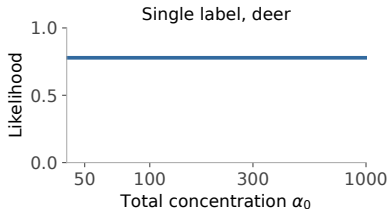
$$p(D_{\text{count}} \mid w) = \prod_{i=1}^N \left[\frac{R_i!}{\prod_{c=1}^C n_{ic}!} \frac{\Gamma(\alpha_0(x_i; w))}{\Gamma(\alpha_0(x_i; w) + R_i)} \prod_{c=1}^C \frac{\Gamma(\alpha_c(x_i; w) + n_{ic})}{\Gamma(\alpha_c(x_i; w))} \right].$$

- Likelihood **can inform total concentration**

- ▶ Does favour one total concentration over another, at fixed mean vectors.
- ▶ Count data can therefore inform aleatoric-distributional allocation.

Fixed mean
 $m_{\text{deer}} = 0.778$

Observed counts
39 deer, 5 ship
7 other labels
 $R = 51$



- **Multinomial BNN** (M-BNN) provides count prediction baseline,

$$p(D_{\text{count}} \mid w) = \prod_{i=1}^N \text{Mult}(n_i \mid R_i, f(x_i; w)).$$

- ▶ Network outputs class probability vector $f(x; w)$.

Model	Predictive uncertainty	Aleatoric share (%)	Distributional share (%)	Epistemic share (%)
M-BNN	0.520 (0.503–0.532)	95.25	–	4.75
DM-eBNN	0.562 (0.554–0.573)	88.28	4.07	7.65

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- DM-eBNN can decompose into all **three uncertainty terms**.
- Counts **can inform** the aleatoric–distributional allocation.
- Does data driven decomposition support **better decisions**?

- Beyond uncertainty decomposition, evaluate **count prediction**.

model	majority label accuracy	count NLL
CIFAR-10H		
M-BNN	0.746 (0.722–0.776)	27.540 (23.630–31.685)
DM-eBNN	0.773 (0.759–0.781)	16.152 (15.664–16.860)

- Beyond uncertainty decomposition, evaluate **count prediction**.

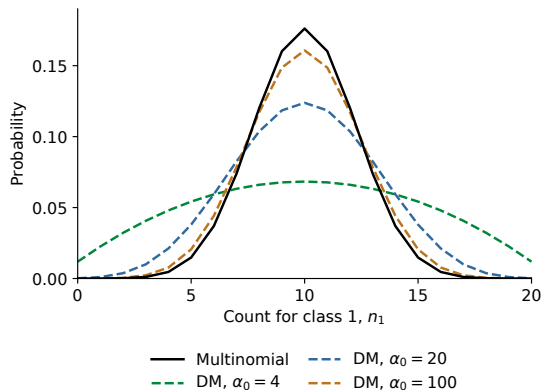
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- DM-eBNN **improves predictive fit**.
- Why?

What Could Explain Predictive Fit Improvement?

- Dirichlet multinomial is more flexible.
- Set class probabilities equal to q .
 - ▶ Count total $R > 1$.
- **Expected counts:** $\mathbb{E}[n_c] = Rq_c$.
- **Variance:** $\text{Var}_{\text{DM}}[n_c] = \frac{\alpha_0 + R}{\alpha_0 + 1} \text{Var}_{\text{Mult}}[n_c]$.
 - ▶ Total concentration controls additional count variance

Two classes, $q = (0.5, 0.5)$, $R = 20$



- **Additional count variance** may explain the improved predictive fit.

- DM-eBNN represents all three uncertainty terms.
- At fixed means, count labels **can** inform total concentration.
 - ▶ Data informed uncertainty allocation.
- Improves held out count prediction in experiments.
- Better decisions from the uncertainty decomposition remain untested.

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- **EDL restricts concentrations to $\alpha_c(x; w) > 1$.**
 - ▶ Test direct positive concentrations, $\alpha_c(x; w) > 0$.
- **Practical value of uncertainty decomposition remains untested.**
 - ▶ Test whether the separate uncertainty terms improve decisions.
- **Training setup**
 - ▶ Multiple chains, larger sampling budgets, learning rate schedules and alternative samplers.
- **Repeated label experiments cover only CIFAR-10H.**
 - ▶ Obtain and evaluate further datasets with repeated labels.

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- Cat-eBNN and DM-eBNN represent **all three uncertainty terms**.
- At fixed means, **total concentration** controls the aleatoric–distributional allocation.
 - ▶ Single labels cannot inform it. Requires prior.
 - ▶ Repeated labels can inform it.
- DM-eBNN improves **held out count prediction** over M-BNN on CIFAR-10H.
- These findings motivate **collecting and retaining repeated labels** for uncertainty-aware classification.

Thank you for your attention :)

- For input x and weights w , $f(x; w)$ lists probabilities for C classes.
- y is the label. $\mathbb{I}[y = c]$ is 1 when $y = c$ and 0 otherwise.

$$p(y \mid x, w) = \text{Cat}(y \mid f(x; w)) = \prod_{c=1}^C f_c(x; w)^{\mathbb{I}[y=c]} = f_y(x; w).$$

- D contains N labelled pairs (x_i, y_i) .
- Conditional independence gives the likelihood as the product over observations.

$$p(D \mid w) = \prod_{i=1}^N p(y_i \mid x_i, w) = \prod_{i=1}^N f_{y_i}(x_i; w).$$

- For input x , class probabilities $q = (q_1, \dots, q_C)$ satisfy $q_c \geq 0$, $\sum_c q_c = 1$.
- Assume R independent categorical labels drawn with probabilities q .
 - ▶ Their counts are $n = (n_1, \dots, n_C)$, with $n_c \in \{0, 1, \dots\}$ and $\sum_c n_c = R$.
- The count vector has a **multinomial distribution**.

$$p(n \mid R, q) = \text{Mult}(n \mid R, q) = \frac{R!}{\prod_{c=1}^C n_c!} \prod_{c=1}^C q_c^{n_c}.$$

- For $R = 1$, label y gives $n_y = 1$ and all other counts zero.
 - ▶ $p(n \mid 1, q) = q_y = \text{Cat}(y \mid q)$.

- For input x , random class probabilities π are **Dirichlet distributed**.

$$p(\pi \mid x, \alpha) = \text{Dir}(\pi \mid \alpha), \quad \alpha_c > 0, \quad \alpha_0 = \sum_{c=1}^C \alpha_c.$$

- Given π , the R labels are independent and their counts are **multinomial**,

$$p(n \mid R, \pi) = \text{Mult}(n \mid R, \pi).$$

- Marginalise** π to obtain count probability at input x .

$$\begin{aligned} p(n \mid x, R, \alpha) &= \text{DM}(n \mid R, \alpha) \\ &= \int_{\Delta^{C-1}} \text{Mult}(n \mid R, \pi) \text{Dir}(\pi \mid \alpha) d\pi \\ &= \frac{R!}{\prod_{c=1}^C n_c!} \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + R)} \prod_{c=1}^C \frac{\Gamma(\alpha_c + n_c)}{\Gamma(\alpha_c)}. \end{aligned}$$

- A (0.5, 0.5) bird–frog class probability prediction can reflect
 - ▶ **Little evidence** for either class.
 - ▶ **Evidence** that supports both classes equally.
- For C classes, subjective logic represents the model's belief about the class label using
 - ▶ b_c , belief assigned to class c .
 - ▶ u , belief not assigned to any individual class.

$$b_c \geq 0, \quad u \geq 0, \quad \sum_{c=1}^C b_c + u = 1.$$

- ▶ a_c , base probability is fraction of unassigned belief allocated to class c

$$a_c \geq 0, \quad \sum_{c=1}^C a_c = 1.$$

- Probability of class c is a function of its **assigned** and **unassigned** belief,

$$p(y = c) = \underbrace{b_c}_{\text{assigned belief}} + \underbrace{a_c u}_{\text{share of unassigned belief}}$$

- Consider $C = 2$ classes, bird ($c = 1$) and frog ($c = 2$).
 - ▶ **Base probabilities** a_c distribute unassigned belief. Choose $a_1 = a_2 = 0.5$.
- **No assigned belief**
 - ▶ **Class belief** b_c assigned to class c . No belief assigned $b_1 = b_2 = 0$.
 - ▶ **Unassigned belief** $u = 1$ (since $\sum_c b_c + u = 1$).

$$p(y = c) = b_c + a_c u = 0 + 0.5 \times 1 = 0.5, \quad c \in \{1, 2\}.$$

- ▶ $(0.5, 0.5)$ predicted class probability reflects **no evidence supporting belief for either class**.
 - **Belief assigned**
 - ▶ Suppose **evidence supports assigning belief** to each class. $b_1 = b_2 = 0.5$ and $u = 0$.
- $$p(y = c) = b_c + a_c u = 0.5 + 0.5 \times 0 = 0.5, \quad c \in \{1, 2\}.$$
- ▶ $(0.5, 0.5)$ predicted class probability reflects **evidence supporting belief for equal classes**.

- The network outputs evidence $e_c(x; w) \geq 0$, **learned support for class c** .
- EDL chooses a **uniform Dirichlet distribution at zero evidence**.
 - ▶ Baseline concentration of one for every class.
- Learned evidence adds to this baseline,

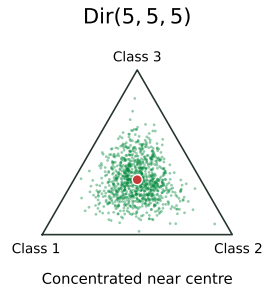
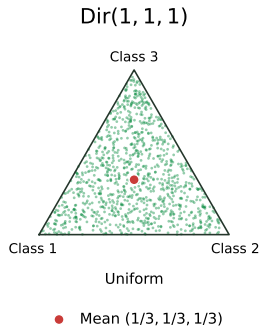
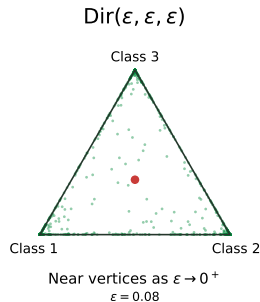
$$\alpha_c(x; w) = \underbrace{1}_{\text{baseline}} + \underbrace{e_c(x; w)}_{\text{learned class evidence}}.$$

- Concentration parameters specify a distribution over random class probability vectors π ,

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)).$$

The Dirichlet Baseline at Zero Evidence

- Each dot is a sampled class-probability vector.
 - Vertices assign probability one to a class.
 - The centre assigns $1/3$ probability to each class.



- EDL chooses **uniform density at zero evidence**

- Following Sensoy et al. (2018) normalised evidence gives **assigned class belief**, normalised baseline gives **unassigned belief**.
 - ▶ Normalise by $\alpha_0(x; w) = \sum_c \alpha_c(x; w)$ gives belief masses,

$$b_c(x; w) = \frac{e_c(x; w)}{\alpha_0(x; w)},$$

$$u(x; w) = 1 - \sum_{c=1}^C b_c(x; w) = \frac{C}{\alpha_0(x; w)}.$$

- ▶ $a_c = 1/C$ base probabilities $\forall c$ since each class has one baseline unit.
- Gives **subjective logic interpretation** of Dirichlet–categorical observation model,

$$\underbrace{p(y = c \mid x, w)}_{\text{observation model}} = \underbrace{b_c(x; w) + a_c u(x; w)}_{\text{subjective logic}} = \underbrace{\frac{\alpha_c(x; w)}{\alpha_0(x; w)}}_{\text{Dirichlet mean}}.$$

- **Entropy** measures label uncertainty,

$$\mathcal{H}[p(y)] = - \sum_{c=1}^C p(y=c) \log p(y=c).$$

- **Expected conditional entropy** averages what remains when the random variable ξ is known,

$$\mathcal{H}[y \mid \xi] = \mathbb{E}_{p(\xi)}[\mathcal{H}[p(y \mid \xi)]] .$$

- **Mutual information** is the average reduction from knowing ξ ,

$$\begin{aligned} \mathcal{I}(y; \xi) &= \mathcal{H}[p(y)] - \mathcal{H}[y \mid \xi], \\ \underbrace{\mathcal{H}[p(y)]}_{\text{total}} &= \underbrace{\mathcal{H}[y \mid \xi]}_{\text{average remaining}} + \underbrace{\mathcal{I}(y; \xi)}_{\text{average reduction}}. \end{aligned}$$

- Apply the decomposition to weights w , then to class probabilities π given w .
- Given input x and training data D ,

$$\underbrace{\mathcal{H}[p(y \mid x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w|D)} [\mathbb{E}_{p(\pi|x,w)} [\mathcal{H}[\text{Cat}(y \mid \pi)]]]}_{\text{aleatoric}} \\ + \underbrace{\mathbb{E}_{p(w|D)} [\mathcal{I}(y; \pi \mid w)]}_{\text{distributional}} \\ + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

- **Categorical BNN** Given x, w , $p(\pi = f(x; w) | x, w) = 1$.

Distributional term is zero.

$$\underbrace{\mathcal{H}[p(y | x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w|D)}[\mathcal{H}[\text{Cat}(y | f(x; w))]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

- **Deterministic Dirichlet network** The weights are fixed at $\hat{w}$.

Epistemic term is zero.

$$\underbrace{\mathcal{H}[p(y | x, \hat{w})]}_{\text{predictive}} = \underbrace{\mathbb{E}_{\text{Dir}(\pi | \alpha(x; \hat{w}))}[\mathcal{H}[\text{Cat}(y | \pi)]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; \pi | \hat{w})}_{\text{distributional}}.$$

- Combining a **weight posterior** with a **Dirichlet distribution** for class probabilities allows all three terms.

- $p(\pi \mid \alpha) = \text{Dir}(\pi \mid \alpha)$, $p(y \mid \pi) = \text{Cat}(y \mid \pi)$, $\alpha_c > 0$.

- Marginalising π gives $p(y \mid \alpha) = m_y = \text{Cat}(y \mid m)$

► $m = \alpha / \alpha_0$, $\alpha_0 = \sum_c \alpha_c$

- Predictive uncertainty decomposes as

$$\underbrace{\mathcal{H}[\text{Cat}(y \mid m)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{\text{Dir}(\pi \mid \alpha)}[\mathcal{H}[\text{Cat}(y \mid \pi)]]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; \pi)}_{\text{distributional}}.$$

- Closed form aleatoric term with $\alpha = \alpha_0 m$,

$$A_m(\alpha_0) := \mathbb{E}_{\text{Dir}(\pi \mid \alpha_0 m)}[\mathcal{H}[\text{Cat}(y \mid \pi)]] = \sum_{c=1}^C m_c [\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1)],$$

- $\psi(t) = \frac{d}{dt} \log \Gamma(t)$ is the digamma function.

Proposition 2.1. Aleatoric Monotonicity

- Fix Dirichlet mean vector m , $0 < m_c < 1 \forall c$.
- **Proposition:** $A_m(\alpha_0)$ is strictly increasing for $\alpha_0 > 0$.
- **Proof** summary: Differentiate at fixed m .

$$A'_m(\alpha_0) = \sum_{c=1}^C m_c [\psi'(\alpha_0 + 1) - m_c \psi'(\alpha_0 m_c + 1)].$$

- ▶ The function $g(t) = t\psi'(t+1)$ is strictly increasing for $t > 0$ (Li et al. 2008, Thm. 1).
- ▶ Since $0 < \alpha_0 m_c < \alpha_0$, $g(\alpha_0 m_c) < g(\alpha_0)$, so

$$\psi'(\alpha_0 + 1) - m_c \psi'(\alpha_0 m_c + 1) > 0.$$

$$A'_m(\alpha_0) > 0.$$

- Fix Dirichlet mean vector m , $0 < m_c < 1 \forall c$.

$$A_m(\alpha_0) = \sum_{c=1}^C m_c [\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1)].$$

- **Proposition:**

$$\lim_{\alpha_0 \rightarrow 0^+} A_m(\alpha_0) = 0, \quad \lim_{\alpha_0 \rightarrow \infty} A_m(\alpha_0) = \mathcal{H}[\text{Cat}(y \mid m)].$$

- **Proof** summary:

- ▶ As $\alpha_0 \rightarrow 0^+$, continuity gives $\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1) \rightarrow 0$.
- ▶ As $z \rightarrow \infty$, $\psi(z) = \log z + \mathcal{O}(z^{-1})$ (e.g. Abramowitz and Stegun 1972, Eq. 6.4.1) gives $\psi(\alpha_0 + 1) - \psi(\alpha_0 m_c + 1) \rightarrow -\log m_c$.

Corollary 2.3. Distributional Uncertainty

- Fix Dirichlet mean vector m , $0 < m_c < 1 \forall c$.

$$\mathcal{I}(y; \pi) = \mathcal{H}[\text{Cat}(y \mid m)] - A_m(\alpha_0).$$

- **Corollary:** $\mathcal{I}(y; \pi)$ is strictly decreasing for $\alpha_0 > 0$, and

$$\lim_{\alpha_0 \rightarrow 0^+} \mathcal{I}(y; \pi) = \mathcal{H}[\text{Cat}(y \mid m)], \quad \lim_{\alpha_0 \rightarrow \infty} \mathcal{I}(y; \pi) = 0.$$

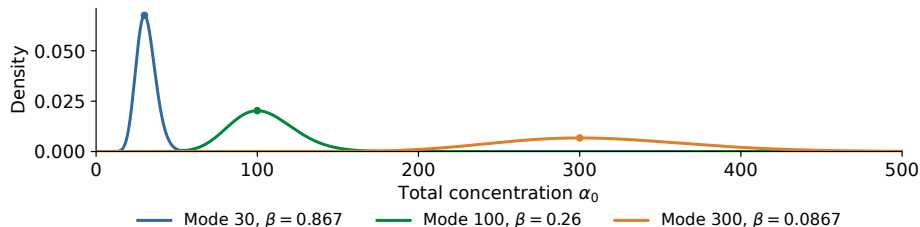
- **Proof** summary:

- ▶ At fixed m , $\mathcal{H}[\text{Cat}(y \mid m)]$ is constant. Proposition 2.1 gives $A_m(\alpha_0)$ strictly increasing, so $\mathcal{I}(y; \pi)$ strictly decreasing.
- ▶ Proposition 2.2 gives $A_m(\alpha_0) \rightarrow 0$ as $\alpha_0 \rightarrow 0^+$ and $A_m(\alpha_0) \rightarrow \mathcal{H}[\text{Cat}(y \mid m)]$ as $\alpha_0 \rightarrow \infty$. Subtracting these limits from $\mathcal{H}[\text{Cat}(y \mid m)]$ gives limits for $\mathcal{I}(y; \pi)$.

$$\text{Gamma}(t \mid \kappa, \beta) = \frac{\beta^\kappa}{\Gamma(\kappa)} t^{\kappa-1} e^{-\beta t}, \quad t > 0, \quad \kappa > 0, \quad \beta > 0,$$

$$\text{Mode} = \frac{\kappa - 1}{\beta} \quad (\kappa > 1), \quad \text{SD} = \frac{\sqrt{\kappa}}{\beta}.$$

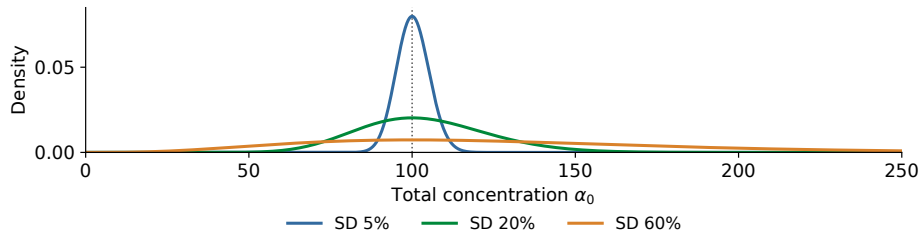
- **Fix shape** $\kappa = 27$, so SD stays approximately 20% of the prior mode.
- **Vary rate** $\beta = 26/\bar{\alpha}_0$ to set prior modes $\bar{\alpha}_0 \in \{30, 100, 300\}$.



$$\text{Gamma}(t \mid \kappa, \beta) = \frac{\beta^\kappa}{\Gamma(\kappa)} t^{\kappa-1} e^{-\beta t}, \quad t > 0, \quad \kappa > 0, \quad \beta > 0,$$

$$\text{Mode} = \frac{\kappa - 1}{\beta} \quad (\kappa > 1), \quad \text{SD} = \frac{\sqrt{\kappa}}{\beta}.$$

- **Fix prior mode** $\bar{\alpha}_0 = 100$, so $\beta = (\kappa - 1)/100$.
- **Vary shape and rate** to set SD to 5%, 20% and 60% of the prior mode.



- For $C = 10$, the EDL parameterisation gives

$$\alpha_0(x; w) = 10 + \sum_{c=1}^{10} e_c(x; w) > 10.$$

- Approaching the lower bound forces **uniform predicted class probabilities**.

$$\alpha_0(x; w) \rightarrow 10^+ \implies \alpha_c(x; w) \rightarrow 1 \implies m_c(x; w) \rightarrow \frac{1}{10} \quad \forall c.$$

- **Fashion-MNIST**, ten classes.
 - ▶ 54000 training, 6000 validation, 10000 test images.
- **CIFAR-10**, ten classes.
 - ▶ 45000 training, 5000 validation, 10000 test images.
- **CIFAR-10H**, repeated labels for the CIFAR-10 test images.
 - ▶ Stratified split of 8000 training, 1000 validation, 1000 test images.
 - ▶ 47–63 labels per image, mean 51.1. Aggregate by class.

- ResNet-20, filter response normalisation.
- Training augmentation: random crops, horizontal flips and per-channel normalisation.
- Three seeds.
- One SGLD chain per seed.

	Fashion-MNIST	CIFAR-10	CIFAR-10H
Cold SGLD epochs	150	600	3000
Deterministic pretraining epochs	50	300	400
Warm SGLD epochs	100	300	1000
Constant SGLD learning rate	0.01	0.01	M-BNN 0.0002 DM-eBNN 0.0012
Retained weight samples	10	10	40
Sample spacing in epochs	5	20	20

- M evaluation inputs. S retained weight samples $w^{(s)}$.

- **Probability of the observed label**

$$p(y_i | x_i, w^{(s)}) = \begin{cases} f_{y_i}(x_i; w^{(s)}), & \text{Softmax, Cat-BNN,} \\ m_{y_i}(x_i; w^{(s)}), & \text{EDL, Cat-eBNN.} \end{cases}$$

- **Negative log likelihood**, averaging probabilities before taking logarithms.

$$\text{NLL} = -\frac{1}{M} \sum_{i=1}^M \log \left[\frac{1}{S} \sum_{s=1}^S p(y_i | x_i, w^{(s)}) \right].$$

- Deterministic models use $S = 1$ and $w^{(1)} = \hat{w}$.

- M evaluation inputs. S weight samples. Count total $R_i = \sum_c n_{ic}$.

- **Probability of the observed count vector**

$$p(n_i \mid x_i, R_i, w^{(s)}) = \begin{cases} \text{Mult}(n_i \mid R_i, f(x_i; w^{(s)})), & \text{M-BNN}, \\ \text{DM}(n_i \mid R_i, \alpha(x_i; w^{(s)})), & \text{DM-eBNN}, \\ \text{Mult}(n_i \mid R_i, m(x_i; w^{(s)})), & \text{mean matched control.} \end{cases}$$

- **Count NLL**, using the corresponding probability above.

$$\text{NLL}_{\text{count}} = -\frac{1}{M} \sum_{i=1}^M \log \left[\frac{1}{S} \sum_{s=1}^S p(n_i \mid x_i, R_i, w^{(s)}) \right].$$

- The control retains DM-eBNN mean vectors and weight samples. No retraining.

- **Cold initialisation** uses random weights.
- **Warm initialisation** loads the best validation checkpoint,
 - ▶ Cat-BNN from Softmax,
 - ▶ Cat-eBNN from EDL,
 - ▶ Count models use matching count likelihoods.
- **Shifted initialisation** adjusts the warm EDL output biases.
 - ▶ Match the initial median total concentration of training subset to the prior mode.
 - ▶ Predicted classes are preserved.
 - ▶ Class probabilities can change.

- Add b to every output bias of w_{warm} , giving weights w_b .

$$\alpha_0(x; w_b) = \sum_{c=1}^C \text{softplus}(z_c(x; w_{\text{warm}}) + b) + C.$$

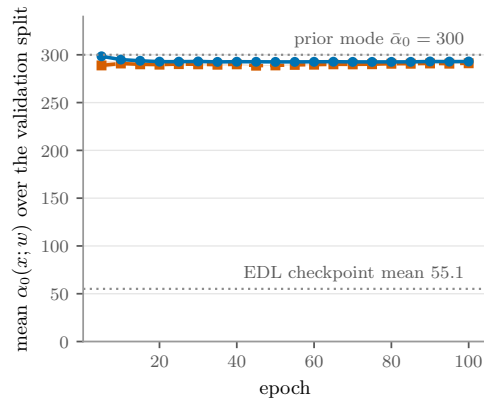
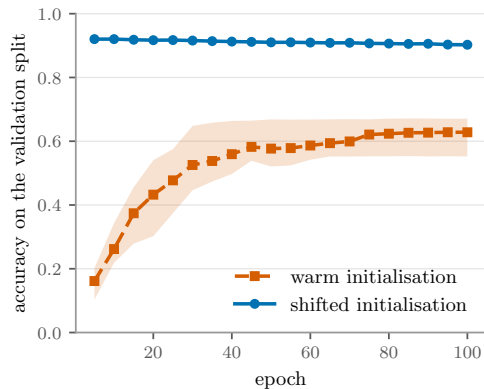
- Match median over subset of training inputs P to the prior mode $\bar{\alpha}_0$.

$$\text{median}_{x \in P} \left[\sum_{c=1}^C \text{softplus}(z_c(x; w_{\text{warm}}) + b) + C \right] = \bar{\alpha}_0.$$

- Solve for b with **Bisection**: b_{low} , b_{high} and tests $b_{\text{mid}} = (b_{\text{low}} + b_{\text{high}})/2$.
 - ▶ Median concentration below $\bar{\alpha}_0 \Rightarrow$ set $b_{\text{low}} = b_{\text{mid}}$.
 - ▶ Median concentration above $\bar{\alpha}_0 \Rightarrow$ set $b_{\text{high}} = b_{\text{mid}}$.
 - ▶ Repeat until close to $\bar{\alpha}_0$.
 - ▶ Apply the final shift b once at initialisation before starting SGLD.

Experimental Setup: Shifted vs Warm Initialisation Diagnostics

- Initialisation affects predictive performance within budget.



BMA accuracy by initialisation method, validation split				
model	Gamma prior mode	cold	warm	shifted
Fashion-MNIST				
Cat-BNN	–	0.871 (0.868–0.874)	0.918 (0.913–0.926)	–
Cat-eBNN	30	0.689 (0.654–0.707)	0.920 (0.915–0.925)	0.855 (0.853–0.856)
Cat-eBNN	58	0.477 (0.472–0.481)	0.899 (0.856–0.924)	0.920 (0.916–0.924)
Cat-eBNN	100	0.230 (0.200–0.277)	0.875 (0.854–0.917)	0.916 (0.912–0.920)
Cat-eBNN	300	0.152 (0.116–0.180)	0.616 (0.553–0.671)	0.908 (0.905–0.910)
CIFAR-10				
Cat-BNN	–	0.668 (0.658–0.680)	0.856 (0.850–0.860)	–
Cat-eBNN	30	omitted	0.839 (0.790–0.863)	0.794 (0.767–0.831)
Cat-eBNN	100	0.231 (0.225–0.239)	0.866 (0.865–0.868)	0.866 (0.864–0.870)
Cat-eBNN	300	omitted	0.831 (0.779–0.857)	0.856 (0.854–0.857)

- Prior SD is 20% of the mode.
- Bold mark selected initialisations.
- **Warm and shifted initialisation** outperform cold initialisation within tested budget.

Single-label predictive performance, test split		
model	accuracy	NLL
Fashion-MNIST		
Softmax	0.928 (0.923–0.933)	0.202 (0.190–0.214)
EDL	0.924 (0.919–0.926)	0.496 (0.477–0.520)
Cat-BNN	0.921 (0.917–0.924)	0.221 (0.211–0.230)
Cat-eBNN	0.913 (0.908–0.916)	0.431 (0.410–0.458)
CIFAR-10		
Softmax	0.879 (0.875–0.882)	0.554 (0.512–0.595)
EDL	0.885 (0.884–0.887)	0.530 (0.517–0.544)
Cat-BNN	0.856 (0.850–0.859)	0.426 (0.418–0.441)
Cat-eBNN	0.867 (0.866–0.868)	0.470 (0.461–0.476)

- Cat-eBNN uses shifted initialisation, prior mode 100 and prior SD 20% of the mode.
- Cat-BNN uses warm initialisation.
- Deterministic models use best validation checkpoint.
- Cat-eBNN achieves **competitive accuracy**.
- **Lower NLL than EDL, but higher NLL than Cat-BNN.**

Gamma prior mode experiment at standard deviation 20%, test split

Gamma prior mode	realised total concentration	predictive uncertainty	aleatoric uncertainty %	distributional uncertainty %	epistemic uncertainty %	NLL
Fashion-MNIST						
30	30.5 (30.5–30.6)	1.336 (1.333–1.341)	90.36	9.39	0.25	0.535 (0.529–0.545)
58	58.4 (58.4–58.5)	0.934 (0.920–0.958)	92.51	7.07	0.42	0.382 (0.373–0.399)
100	99.0 (98.8–99.3)	0.974 (0.934–1.008)	95.42	4.13	0.45	0.431 (0.410–0.458)
300	292.7 (292.3–292.9)	1.947 (1.922–1.959)	99.12	0.77	0.11	1.143 (1.111–1.168)
CIFAR-10						
30	30.4 (30.2–30.5)	1.414 (1.399–1.436)	89.24	8.98	1.78	0.763 (0.682–0.923)
100	99.7 (99.5–99.8)	0.818 (0.813–0.822)	91.11	4.79	4.10	0.470 (0.461–0.476)
300	291.4 (291.2–291.6)	1.733 (1.728–1.739)	98.07	0.87	1.06	0.916 (0.907–0.921)

- Prior SD is fixed at 20% of the mode.
- Mode 30 uses warm initialisation. Higher modes use shifted initialisation.
- Higher prior modes increase realised total concentration
 - **Shifts allocation** towards aleatoric uncertainty.

Gamma prior standard deviation experiment at mode 100, test split

Gamma prior standard deviation	realised total concentration	predictive uncertainty	aleatoric uncertainty %	distributional uncertainty %	epistemic uncertainty %	NLL
Fashion-MNIST						
1000%	67.6 (66.8–68.8)	0.854 (0.836–0.874)	92.40	7.11	0.49	0.359 (0.348–0.374)
60%	90.4 (89.6–91.3)	0.903 (0.879–0.935)	94.74	4.85	0.41	0.393 (0.378–0.418)
20%	99.0 (98.8–99.3)	0.974 (0.934–1.008)	95.42	4.13	0.45	0.431 (0.410–0.458)
5%	100.1 (100.1–100.2)	1.029 (0.988–1.061)	95.68	3.90	0.42	0.454 (0.433–0.482)
CIFAR-10						
1000%	83.7 (82.2–85.0)	0.815 (0.812–0.818)	89.39	6.03	4.58	0.473 (0.465–0.477)
60%	97.2 (97.1–97.3)	0.800 (0.798–0.803)	90.56	5.03	4.41	0.466 (0.458–0.472)
20%	99.7 (99.5–99.8)	0.818 (0.813–0.822)	91.11	4.79	4.10	0.470 (0.461–0.476)
5%	100.1 (100.0–100.1)	0.919 (0.892–0.934)	92.57	4.32	3.11	0.508 (0.499–0.516)

- Prior mode is fixed at 100. Shifted initialisation.
- Prior SD is varied as a percentage of the mode.
- Narrower priors push total concentration closer to the mode
 - ▶ Resulting in higher total concentration.
 - ▶ **Shifts allocation** towards aleatoric uncertainty.

Predictive uncertainty and its decomposition, test split				
model	predictive uncertainty	aleatoric uncertainty	distributional uncertainty	epistemic uncertainty
Fashion-MNIST				
Softmax	0.183	0.183 (100.00%)	–	–
EDL	1.038	0.926 (89.21%)	0.112 (10.79%)	–
Cat-BNN	0.233	0.225 (96.55%)	–	0.008 (3.45%)
Cat-eBNN	0.974	0.929 (95.42%)	0.040 (4.13%)	0.004 (0.45%)
CIFAR-10				
Softmax	0.110	0.110 (100.00%)	–	–
EDL	0.815	0.732 (89.77%)	0.083 (10.23%)	–
Cat-BNN	0.390	0.318 (81.38%)	–	0.073 (18.62%)
Cat-eBNN	0.818	0.746 (91.11%)	0.039 (4.79%)	0.034 (4.10%)

- Averages over test inputs and three seeds. Parentheses give uncertainty shares.
- Cat-eBNN uses shifted initialisation, prior mode 100 and prior SD 20% of the mode.
- Cat-BNN uses warm initialisation.
- Cat-eBNN represents **all three uncertainty terms**,
 - ▶ Its aleatoric–distributional allocation **depends on the prior assumption**.

model	Gamma prior mode	realised total concentration	Count prediction, test split		
			majority label accuracy	count NLL	mean matched control NLL
Cold initialisation					
M-BNN	–	–	0.724 (0.712–0.744)	19.161 (18.012–19.754)	–
DM-eBNN	50	50.0 (49.6–50.6)	0.712 (0.697–0.731)	12.718 (12.486–13.007)	23.719 (23.195–24.467)
DM-eBNN	200	178.6 (176.4–180.3)	0.702 (0.693–0.715)	13.199 (11.872–14.051)	20.291 (17.602–22.035)
Warm initialisation					
M-BNN	–	–	0.746 (0.722–0.776)	27.540 (23.630–31.685)	–
DM-eBNN	50	51.0 (50.8–51.1)	0.753 (0.720–0.777)	14.392 (13.724–15.529)	28.802 (26.771–32.415)
DM-eBNN	200	182.7 (181.5–184.8)	0.773 (0.759–0.781)	16.152 (15.664–16.860)	26.207 (25.281–27.394)

- DM-eBNN prior SD is 20% of mode.
- Cold SGLD 3000 epochs, warm SGLD 1000 epochs after 400 pretraining.
- DM-eBNN **lower count NLL** than M-BNN and mean matched controls.

initialisation	Gamma prior sensitivity, test split					
	pretraining Gamma prior mode, standard deviation	initial realised total concentration	SGLD Gamma prior mode, standard deviation	realised total concentration	majority label accuracy	count NLL
cold	none	–	none	10.0 (10.0–10.0)	0.099 (0.098–0.100)	23.433 (23.433–23.433)
warm	50, 20%	52.1 (51.6–52.6)	50, 20%	51.0 (50.8–51.1)	0.753 (0.720–0.777)	14.392 (13.724–15.529)
warm	50, 20%	52.1 (51.6–52.6)	50, 100%	69.8 (69.2–70.3)	0.754 (0.723–0.776)	13.626 (12.863–14.940)
warm	50, 20%	52.1 (51.6–52.6)	none	79.3 (78.2–80.1)	0.752 (0.724–0.776)	13.657 (12.863–15.032)
warm	200, 20%	186.7 (184.4–189.9)	200, 20%	182.7 (181.5–184.8)	0.773 (0.759–0.781)	16.152 (15.664–16.860)
warm	200, 20%	186.7 (184.4–189.9)	200, 100%	162.5 (160.6–165.5)	0.771 (0.758–0.781)	14.439 (13.983–15.153)
warm	200, 20%	186.7 (184.4–189.9)	none	155.9 (154.1–159.0)	0.772 (0.760–0.784)	13.941 (13.557–14.574)

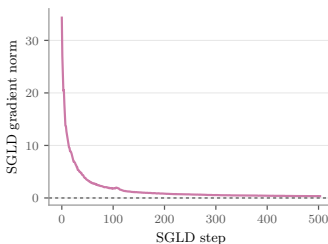
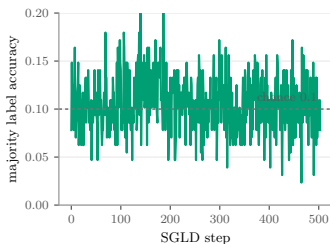
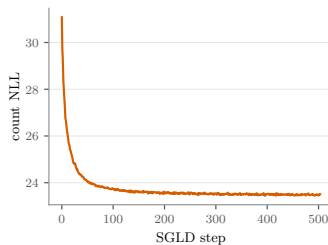
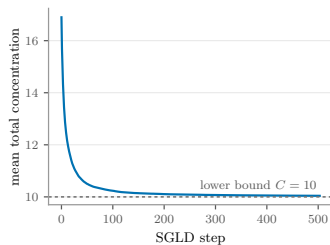
- Warm pretraining uses prior modes 50 or 200, with SD 20% of the mode.
- During SGLD, prior SD is 20% or 100% of the mode, or no Gamma prior.
- Wider or absent priors result in more **data informed total concentration** and improve count NLL.
- Cold training without a Gamma prior collapses.

model	Predictive uncertainty of the class label, test split				
	Gamma prior mode	predictive uncertainty	aleatoric uncertainty %	distributional uncertainty %	epistemic uncertainty %
M-BNN	–	0.520 (0.503–0.532)	95.25	–	4.75
DM-eBNN	50	1.040 (1.036–1.047)	87.60	7.71	4.69
DM-eBNN	200	0.562 (0.554–0.573)	88.28	4.07	7.65

- Averages over test inputs and three seeds. Parentheses give uncertainty shares.
- Warm initialisation.
- DM-eBNN prior modes 50 and 200, with SD 20% of the mode.
- DM-eBNN represents **all three uncertainty** terms,
 - ▶ Counts are **able to inform** the aleatoric–distributional split.

Training without the Gamma Prior

- Count data can inform total concentration. Can we remove Gamma prior?



- Early updates lower count NLL by reducing total concentration.
 - As $\alpha_0(x; w) \rightarrow 10$, mean vectors become uniform.
 - Accuracy stays near chance.
- Evidence approaches zero.
 - Softplus derivatives vanish. Gradients become small.